Approximating Self-attention

Target: deriving new RNN states by approximating softmax attention

🍉 Key tools

1. First-order Taylor Expansion $f(x') = f(x + \Delta x) = f(x) + \nabla_x f(x)|_x \cdot \Delta x$
2. Approximation of exponentials
   a. $e^x \approx 1 + x$ near 0
   b. The performer way: $e^{k^T q} = \mathbf{E}_{\omega \in \mathcal{N}(0, \mathbf{I}_d)} e^{k^T w - \|k\|_2/2} \cdot e^{q^T w - \|q\|_2/2}$

🍉 Observations

1. Approximating the exponential => building RNN states.
2. Taylor expansion (of normalizers) => delta rules.

Denote the self-attention output (or correspondingly, readout vector of RNN),

$$o_q^i = \sum_{j=1}^{i} \frac{\exp\{k_j^T q\} v_j}{\sum_{l=1}^{i} \exp\{k_l^T q\}} = \sum_{j=1}^{i} \frac{\exp\{k_j^T q\}}{Z_q^i} v_j = \sum_{j=1}^{i} P_q(j) v_j = \mathbf{E}_{P_q}[v] \tag{1}$$

Now, let's consider changes from step i to i+1. We always write $q \triangleq q_i, \quad q' \triangleq q_{i+1}, \quad \Delta q \triangleq q' - q$,

$$o_{q'}^{i+1} = \sum_{j=1}^{i+1} \frac{\exp\{k_j^T q'\}}{Z_{q'}^{i+1}} v_j = \sum_{j=1}^{i} \frac{\exp\{k_j^T q'\}}{Z_{q'}^{i+1}} v_j + \frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1} \tag{2}$$

$$= \sum_{j=1}^{i} \underbrace{\frac{\exp\{k_j^T q'\} v_j}{Z_{q'}^{i}} \frac{Z_{q'}^{i}}{Z_{q'}^{i+1}}}_{\text{online softmax trick}} + \frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1} \tag{3}$$

$$= \underbrace{o_{q'}^{i}}_{\text{part A}} \underbrace{\frac{Z_{q'}^{i}}{Z_{q'}^{i+1}}}_{\text{part B}} + \underbrace{\frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1}}_{\text{part C}} \tag{4}$$

Part A

For part A, we apply the first-order Taylor expansion,

$$o_{q'}^{i} = o_q^i + \nabla_q o_q^i \cdot \Delta q \tag{5}$$

where we compute the gradient with "log trick" as policy gradients,

$$\nabla_q o_q^i = \nabla_q \mathbf{E}_{P_q}[v] = \nabla_q \sum_{j=1}^{i} P_q(j) v_j = \sum_{j=1}^{i} \nabla_q P_q(j) v_j = \sum_{j=1}^{i} P_q(j) v_j \nabla_q^T \log P_q(j) = \mathbf{E}_{P_q}[v \nabla_q^T \log P_q]($$

$$\nabla_q \log P_q(j) = \nabla_q \log \frac{\exp\{k_j^T q\}}{Z_q^i} = k_j - \nabla_q \log Z_q^i = k_j - \frac{1}{Z_q^i} \nabla_q \sum_{l=1}^{i} \exp\{k_j^T q\} \tag{7}$$

$$= k_j - \sum_{l=1}^{i} \frac{\exp\{k_j^T q\}}{Z_q^i} k_j = k_j - \mathbf{E}_{P_q}[k] \tag{8}$$

Therefore,

$$\nabla_q o_q^i = \sum_{j=1}^{i} P_q(j) v_j (k_j - \mathbf{E}_{P_q}[k])^T = \mathbf{E}_{P_q}[v k^T] - \mathbf{E}_{P_q}[v] \mathbf{E}_{P_q}[k^T] = \mathbf{E}_{P_q}[v k^T] - o_q^i \mathbf{E}_{P_q}[k^T] \tag{9}$$

$$o_{q'}^{i} = o_q^i + \nabla_q o_q^i \cdot \Delta q + o(\|\Delta q\|) \tag{10}$$

$$\approx o_q^i + \mathbf{E}_{P_q}[v k^T] \Delta q - o_q^i \mathbf{E}_{P_q}[k^T] \Delta q \tag{11}$$

Since the expectation $\mathbf{E}_{P_q}$ still needs to run over sequences, we can further approximate P_q with a uniform distribution $P_q(j) = 1/i$,

> - The approximation here may be improved.

> - In policy gradient, the expectation is usually approximated by sampling, that is, $\mathbf{E}_{P_q} v k^T = \frac{1}{m} \sum_{s=1}^{m} v_s k_s^T$
> - We may take the policy gradient view and go for the test-time training formulation. That is, as

$$\mathbf{E}_{P_q} v k^T \approx \frac{1}{i} \sum_{j=1}^{i} v_j k_j^T \triangleq S_i, \quad \mathbf{E}_{P_q} k^T \approx \frac{1}{i} \sum_{j=1}^{i} k_j^T \triangleq K_i^T \tag{12}$$

We have,

$$\begin{aligned} o_{q'}^i &\approx o_q^i + \mathbf{E}_{P_q}[v k^T] \Delta q - o_q^i \mathbf{E}_{P_q}[k^T] \Delta q & (13) \\ &\approx o_q^i + S_i \Delta q - o_q^i K_i^T \Delta q & (14) \\ &= o_q^i + \underbrace{(S_i - o_q^i K_i^T)}_{\text{delta rule}} \Delta q & (15) \end{aligned}$$

> Remarks on alternative approximations (those applied in linear attention)
>
> 1. We derive a delta rule by taking normalizor Z_q^i in the Taylor expansion,
>
> **Vanilla Linear Attention:**
>
> $$\text{Parallel training:} \quad \boldsymbol{O} = \cancel{\text{softmax}}\left(\boldsymbol{Q}\boldsymbol{K}^\top + \log \boldsymbol{M}\right)\boldsymbol{V} = \left(\boldsymbol{Q}\boldsymbol{K}^\top \odot \boldsymbol{M}\right)\boldsymbol{V} \quad \in \mathbb{R}^{n\times}$$
>
> $$\text{Iterative inference:} \quad \boldsymbol{o}_t = \sum_{j=1}^{t} \frac{\cancel{\exp}\left(\boldsymbol{q}_t^\top \boldsymbol{k}_j\right)}{\cancel{\sum_{l=1}^{t} \exp\left(\boldsymbol{q}_t^\top \boldsymbol{k}_l\right)}} \boldsymbol{v}_j \quad = \sum_{j=1}^{t} \left(\boldsymbol{q}_t^\top \boldsymbol{k}_j\right) \boldsymbol{v}_j \quad \in \mathbb{R}^d$$
>
> $\boldsymbol{M}$ 改为线性注意力的causal mask： $\boldsymbol{M}_{i,j} = \begin{cases} 0 & \text{if } j > i \\ 1 & \text{if } j \le i \end{cases}$
>
> **RNN-like inference:**
>
> $$\begin{aligned} \boldsymbol{o}_t &= \sum_{j=1}^{t} \boldsymbol{v}_j \left(\boldsymbol{k}_j^\top \boldsymbol{q}_t\right) & \boldsymbol{k}_j^\top \boldsymbol{q}_t = \boldsymbol{q}_t^\top \boldsymbol{k}_j \in \mathbb{R} \\ &= \left(\sum_{j=1}^{t} \boldsymbol{v}_j \boldsymbol{k}_j^\top\right) \boldsymbol{q}_t & \text{By associativity} \end{aligned}$$
>
> 定义状态 S_t， 重写线性注意力形式：
>
> $$\boldsymbol{S}_t = \sum_{j=1}^{t} \boldsymbol{v}_j \boldsymbol{k}_j^\top = \boldsymbol{S}_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^\top \in \mathbb{R}^{d\times d}, \quad \boldsymbol{o}_t = \boldsymbol{S}_t \boldsymbol{q}_t \in \mathbb{R}^d$$
>
> 2. By performing Tayler expansion at q , instead at 0 , we derive a new state (see discussions on part B/C).

Part B/C

In parts B and C, we also approximate $Z_{q'}^i$ through a first-order Taylor expansion

$$
\begin{aligned}
Z_{q'}^i &= \sum_{j=1}^{i} \exp\{k_j^T q'\} = \sum_{j=1}^{i} \exp\{k_j^T q\} + \nabla_q \sum_{j=1}^{i} \exp\{k_j^T q\} \cdot \Delta q + o(\|\Delta q\|) && (16) \\
&\approx Z_q^i + \sum_{j=1}^{i} \exp\{k_j^T q\} k_j^T \Delta q && (17) \\
&\approx Z_q^i + \sum_{j=1}^{i} (k_j^T q + 1) k_j^T \Delta q \quad \text{// assume } k_j^T q \text{ nears 0, may not be accurate in the sense of approxiation} && (18) \\
&\approx Z_q^i + q^T \left(\sum_{j=1}^{i} k_j k_j^T \right) \Delta q + \left(\sum_{j=1}^{i} k_j^T \right) \Delta q && (19) \\
&= Z_q^i + q^T X_i \Delta q + K_i^T \Delta q && (20)
\end{aligned}
$$

Where $X_i \triangleq \sum_{j=1}^{i} k_j k_j^T$, and we have

$$
\text{part B: } \frac{Z_{q'}^i}{Z_{q'}^{i+1}} = \frac{Z_{q'}^i}{Z_{q'}^i + \exp\{k_i^T q'\}} \tag{21}
$$

$$
\begin{aligned}
& && (22) \\
\text{update states: } Z_{q'}^{i+1} &= Z_{q'}^i + \exp\{k_i^T q'\} && (23) \\
&= Z_q^i + q^T X_i \Delta q + K_i^T \Delta q + \exp\{k_{i+1}^T q'\} && (24) \\
X_{i+1} &= X_i + k_{i+1} k_{i+1}^T && (25)
\end{aligned}
$$

Part C can be obtained from $Z_{q'}^{i+1}$

$$
\text{part C: } \frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1} \tag{26}
$$

> Alternatively, we can have a simpler expansion at 0 ,
>
> $$
> Z_{q'}^{i+1} = \sum_{j=1}^{i+1} \exp\{k_j^T q'\} \approx \sum_{j=1}^{i+1} k_j^T q' = K_{i+1}^T q' \tag{27}
> $$
>
> where we only track K_{i+1} in the update, leaving the second-order states X_i untouched.

Full Picture

🍉 Update Rules

$$
\begin{aligned}
\Delta q &= q_{i+1} - q_i = q' - q && (28)\\
o_{q'}^{i+1} &= \underbrace{o_{q'}^{i}}_{\text{part A}} \underbrace{\frac{Z_{q'}^{i}}{Z_{q'}^{i+1}}}_{\text{part B}} + \underbrace{\frac{\exp\{k_i^T q'\}}{Z_{q'}^{i+1}}}_{\text{part C}} v_i && (29)\\
&= \left(o_q^i + (S_i - o_q^i K_i^T)\Delta q\right) \times \left(1 + \frac{\exp\{k_{i+1}^T q'\}}{Z_q^i + q^T X_i \Delta q + K_i^T \Delta q}\right)^{-1} + \frac{\exp\{k_{i+1}^T q'\}}{Z_q^i + q^T X_i \Delta q + K_i^T \Delta q} v_{i+1} && (30)\\
Z_{q'}^{i+1} &= Z_q^i + q^T X_i \Delta q + K_i^T \Delta q + \exp\{k_{i+1}^T q'\} && (31)\\
S_{i+1} &= \frac{i}{i+1} S_i + \frac{1}{i+1} v_{i+1} k_{i+1}^T && (32)\\
K_{i+1} &= \frac{i}{i+1} K_i + \frac{1}{i+1} k_{i+1} && (33)\\
X_{i+1} &= X_i + k_{i+1} k_{i+1}^T && (34)
\end{aligned}
$$

> Linear attentions: the readout function is multiplicative ($o = f(S, q) = Sq$)
>
> Here, the readout is additive $o = f(S, q, v) = Sq + v$

🍉 Refinement 1

$$
\begin{aligned}
\beta_{i+1} &= \frac{\exp\{k_{i+1}^T q'\}}{Z_q^i + q^T X_i \Delta q + K_i^T \Delta q} && (35)\\
o_{q'}^{i+1} &= (1 - \beta_{i+1}) \left(o_q^i + (S_i - o_q^i K_i^T)\Delta q\right) + \beta_{i+1} v_{i+1} \quad // \text{ if } \beta_{i+1} \text{ is small} && (36)\\
Z_{q'}^{i+1} &= \exp\{k_{i+1}^T q'\} / \beta_{i+1} + \exp\{k_{i+1}^T q'\} && (37)
\end{aligned}
$$

Or by the simpler approximation,

$$
\begin{aligned}
\Delta q &= q_{i+1} - q_i = q' - q && (38)\\
\beta_{i+1} &= \frac{\exp\{k_{i+1}^T q'\}}{K_i^T q'} && (39)\\
o_{q'}^{i+1} &= (1 - \beta_{i+1}) \left(o_q^i + (S_i - o_q^i K_i^T)\Delta q\right) + \beta_{i+1} v_{i+1} \quad // \text{ if } \beta_{i+1} \text{ is small} && (40)\\
S_{i+1} &= \frac{i}{i+1} S_i + \frac{1}{i+1} v_{i+1} k_{i+1}^T && (41)\\
K_{i+1} &= \frac{i}{i+1} K_i + \frac{1}{i+1} k_{i+1} && (42)
\end{aligned}
$$